



Extraction and separation of manganese and iron from ferruginous manganese ores: A review

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ABSTRACT

There are abundant ferruginous manganese ores (abbr. Fe-Mn ores) in many parts of the world. For example, almost half of the manganese ore resources in India are ferruginous and more than 73% of the manganese ores in China belong to Fe-Mn ores with a low Mn/Fe mass ratio (< 3). With the depletion of high grade manganese ore resources, the Fe-Mn ores are becoming important substitution resources for extracting manganese. In general, the production of Mn alloy from manganese ores requires that the Mn grade is more than 30% and the Mn/Fe mass ratio is greater than 5. During the production of electrolytic manganese or manganese chemical products, the issue of co-leaching of Mn and Fe complicates the purification procedure for the MnSO_4 solution if Fe-Mn ores are used as raw materials.

Numerous approaches are reported to realize the selective extraction and separation of Mn and Fe from the Fe-Mn ores. The extraction and separation technologies cover physical beneficiation, chemical beneficiation (hydrometallurgy and pyrometallurgy) and physico-chemical combined processes. This present work reviews the technical principles, parameters and recovery efficiencies of diverse processes on the aspect of selective extraction and separation of Mn and Fe from Fe-Mn ores. This review can provide guidance for selecting appropriate methods to exploit Fe-Mn ores or other secondary resources containing Fe and Mn oxides. The authors also put forward a new route to produce manganese ferrite materials using Fe-Mn ores as the raw materials.

1. Introduction

Manganese is an important metal with wide industrial applications in many aspects of social economy. Approximately 90–95% of manganese are consumed in the steel production, and the rest are used in the fields of non-ferrous production, battery, and dietary additives (Tan et al., 2004; Mei et al., 2011; IMnI, 2008–2016; China Manganese Industry). Manganese oxide ores and carbonate ores are two main natural minerals on earth for extraction of manganese. Manganese is mostly found in minerals in association with iron due to similar geochemical properties (Yao, 1996; Fan and Yang, 1999). According to the Chinese industrial classifications of manganese oxide ores and carbonate ores by Mn/Fe mass ratio (Yao, 1996; Fan and Yang, 1999; Tan et al., 2004), manganese ores are divided into three types, namely rich Mn ores (higher Mn grade), Fe-Mn ores and poor Mn ores (lower Mn grade) (as listed in Table 1). In recent years, the world annual demand for manganese has increased significantly due to soaring steel production (USGS, 2006–2016; IMnI, 2008–2016). As rich Mn ores are exhausted gradually, many researchers have paid much attention to extract and recover manganese from Fe-Mn ores and poor Mn ores to

satisfy the development requirement of manganese industry (Yan and Gao, 2009; Liu et al., 2015; Zhu, 2016).

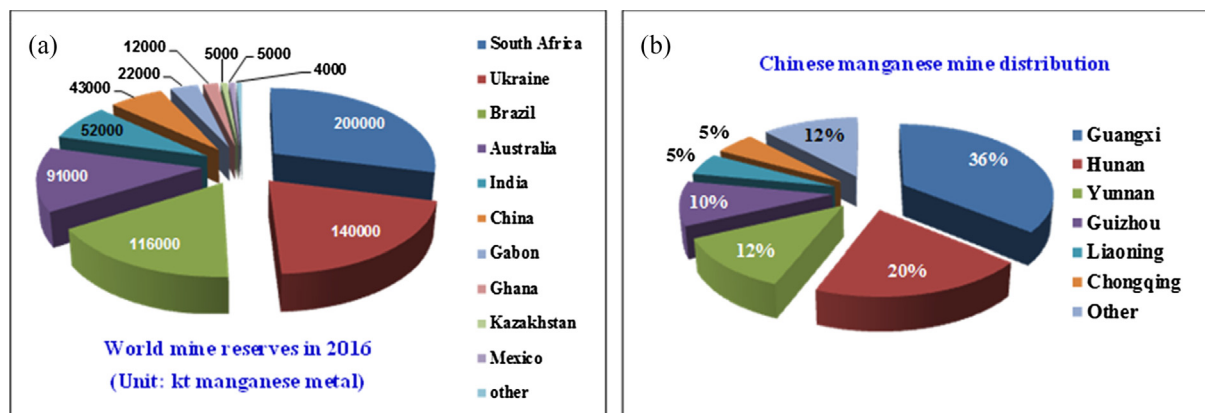
Fe-Mn ores and poor Mn ores resources are abundant all over the world (Yao, 1996; Fan and Yang, 1999; Tan et al., 2004; Granina et al., 2010; Baoumy et al., 2013; El-Hussiny et al., 2015). During the mining, crushing, beneficiation and transportation of rich Mn ores, about 30% of the mined ores are converted into manganese ore fines with low Mn grade (Singh et al., 2011). In India, almost half of manganese ore resources are ferruginous in nature (Singh et al., 2011; Singh and Biswas, 2017). In China, $> 73\%$ of manganese ores belong to Fe-Mn ores with a low Mn/Fe mass ratio (< 3) (Tan et al., 2004; Zhang et al., 2013). A lot of Fe-Mn ores are also found in Egypt (El-Hussiny et al., 2015) and Russia (Granina et al., 2010). In general, the commercial grade of manganese ores for Mn alloys production requires $\text{Mn} > 30\%$ and $\text{Mn/Fe mass ratio} > 5$ (Tan et al., 2004). Many techniques have been tried to separate Mn and Fe from Fe-Mn ores in order to improve the Mn grade and Mn/Fe mass ratio. On the other hand, as for the production of electrolytic manganese or manganese chemical products, co-leaching of Mn and Fe complicates the purification procedure of MnSO_4 solution when the Fe-Mn ores are used as raw materials.

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Table 1

China industrial classifications of manganese ore types by Mn/Fe mass ratio (Yao, 1996; Fan and Yang, 1999; Tan et al., 2004).

Mn ore type	Industrial classification	Grade	Limiting Mn grade (%)	Industrial average Mn grade (%)	Mn + Fe (%)	Mn/Fe ratio	P/Mn ratio	SiO ₂ (%)
Oxide ore	Rich Mn ores	I	35	40	–	≥ 6	≤ 0.004	≤ 15
		II	30	35	–	≥ 4	≤ 0.005	≤ 25
		III	18	30	–	≥ 3	≤ 0.004	≤ 25
	Fe-Mn ores	I	20	25	≥ 50	< 3	≤ 0.2	≤ 25
		II	15	20	≥ 40	< 3	≤ 0.2	≤ 25
		III	10	15	≥ 30	< 3	≤ 0.2	≤ 35
	Poor Mn ores	–	~ 15	18	–	–	–	–
Carbonate ore	Rich Mn ores	–	15	25	–	≥ 3	< 0.005	≤ 25
	Fe-Mn ores	–	10	15	≥ 25	–	≤ 0.2	≤ 35
	Poor Mn ores	–	10	15	–	–	–	–
	Mn-bearing limestone	–	8	12	–	–	–	–

**Fig. 1.** World manganese mine reserves and Chinese manganese mine distribution (USGS; IMnI; China Manganese Industry).

In order to obtain specification-level Mn-bearing concentrates with higher Mn/Fe mass ratio or manganese solutions with lower content of iron ion, many works have been conducted for selective extraction and separation of Mn and Fe from the Fe-Mn ores. Previous investigations have indicated that it is very difficult to effectively separate Mn and Fe only using physical beneficiation methods, such as gravity, magnetic or flotation processes, owing to closely disseminated relationship among Mn, Fe and Si constituents in the Fe-Mn ores (isomorphic substitution of Fe and Mn elements in the mineral lattice during the natural mineralization of Fe-Mn oxides and silicates) or the sliming effect during the physical separation processes (Singh et al., 2010; Baioumy et al., 2013; Tripathy et al., 2015; X.D. Wu et al., 2015). Although most gangue minerals can be removed easily by physical separation techniques and the Mn grade of the final concentrates is obviously upgraded, the concentrates still contain higher iron components than the specifications needed for the concentrate. Therefore, the obtained concentrates should be further treated by chemical separation methods to separate Mn from Fe. These chemical separation methods include direct reduction leaching (Pagnanelli et al., 2004; El-Hazek et al., 2006; Nayl et al., 2011), Mn-rich slag smelting method (Tan et al., 2004; Mei et al., 2011), carbothermic reduction roasting followed by separation processes (Gao et al., 2012a, 2012b; Zhao, 2012; Cai, 2013; Duan, 2013; X.D. Wu et al., 2015; Ming et al., 2016; Zhang et al., 2017a), sulfation roasting followed by a leaching process (Cai et al., 2012; Zhang et al., 2013; You et al., 2015, 2017).

Up to now, the reduction, leaching and bioleaching of rich Mn ores (Mn > 30%), secondary materials and marine manganese nodules have been reviewed in the literatures concentrating on the recovery of Mn, while ignoring the reduction or leaching characteristics of Fe oxides in the manganese ores (Eriç, 1995; Mukherjee et al., 2004; Zhang and Cheng, 2007a, 2007b, 2007c; Li, 2008; Huang et al., 2013; Cheng et al., 2015; Das et al., 2011). The present review mainly focuses on the physical beneficiation, direct reduction leaching, Mn-rich slag smelting,

carbothermic reduction roasting followed by separation processes and the sulfation roasting followed by leaching processes on the aspect of extraction and separation of Mn and Fe from the Fe-Mn ores.

2. Manganese resources and consumption

Fig. 1 shows the world manganese mine reserves and Chinese manganese mine distribution. Global manganese mineral resources are very rich but distributed unevenly. According to the U.S. Geological Survey (USGS) statistics of 2016, the global manganese ore reserve is about 690 million tons (metal basis), and 99.5% of manganese deposits are concentrated in South Africa, Ukraine, Brazil, Australia, India, China, Gabon, Ghana, Kazakhstan and Mexico. As reported, the rich Mn ores containing more than 35% Mn are mainly distributed in South Africa, Australia, Gabon and Brazil (USGS; IMnI; China Manganese Industry). Although the Ukrainian Mn ores (total tonnage) rank second in the world, 70% of the Mn ores belong to low grade manganese carbonate ores with average Mn grade of 16–19%, and the average Mn grade of manganese oxide ores is only 22–27% (USGS; IMnI; China Manganese Industry; Tan et al., 2004).

The manganese ore resources in China are about 43,000 kilotons (metal basis) and rank the 6th of the world (China Manganese Industry). The distribution of manganese resources is also unbalanced, and 68% of the manganese deposits are distributed in the Guangxi, Hunan and Yunnan provinces, which is also named as the “Manganese triangle” of China. However, the average Mn grade of Chinese manganese resources is only about 20–22%. Moreover, 73% of the manganese resources belong to Fe-Mn ores, and 49.6% of the manganese resources contain a high content of phosphorus (P/Mn > 0.005). Fig. 2 presents the mineralogy and texture of typical Fe-Mn ores in China (Jiangxi province) and South Africa (Northern Cape province). These two kinds of Fe-Mn ores are characterized as oxide type manganese ores. It can be seen from Fig. 2 that the microstructure of the Fe-Mn ores

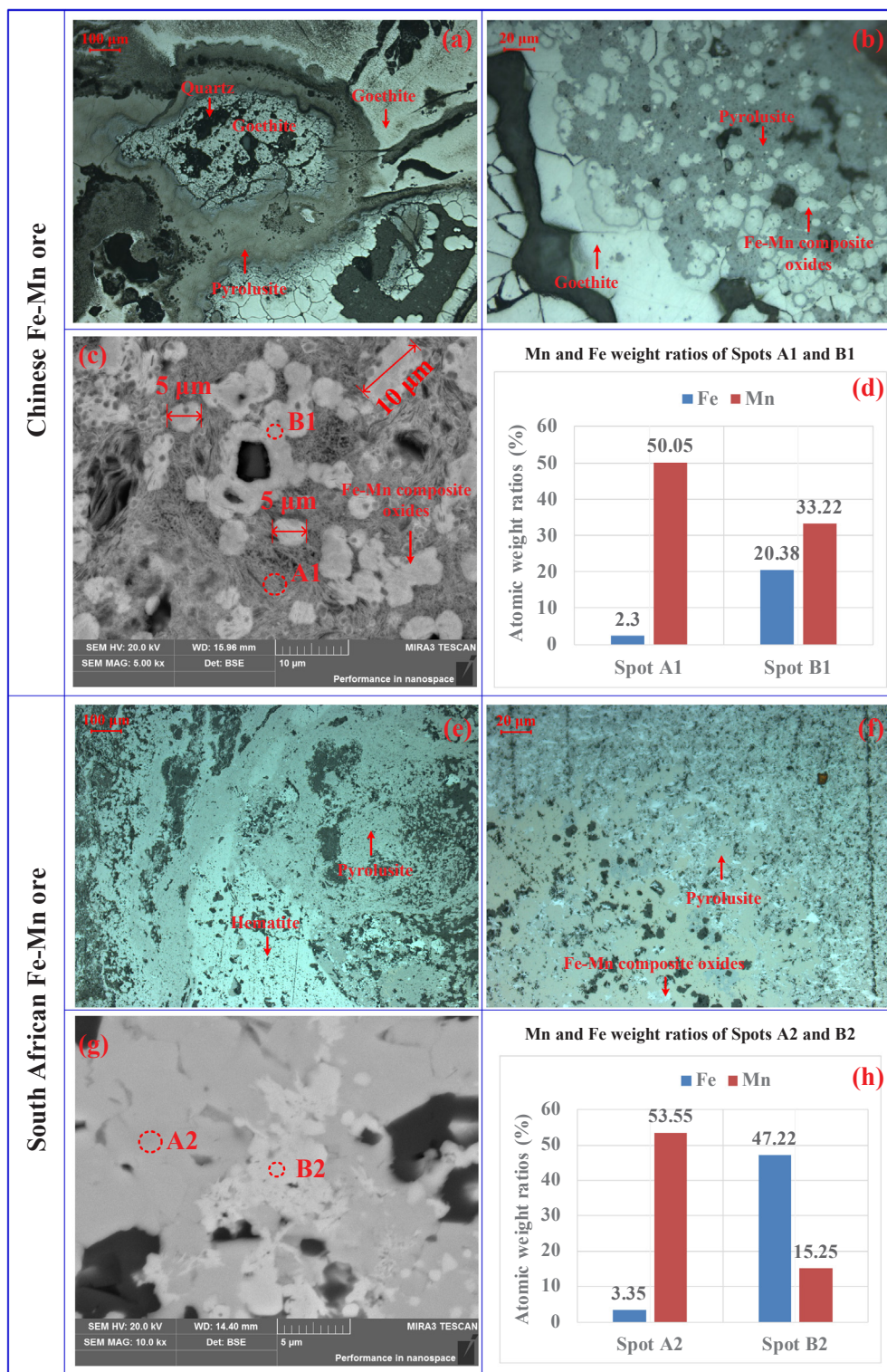


Fig. 2. Optical microscopy and SEM-EDS analyses of typical Fe-Mn ores in China and South Africa, (a) and (b) optical microstructure images with magnification of 100 $\times$; (e) and (f) optical microstructure images with magnification of 500 $\times$; (c) SEM image of the micro-region in (b); (g) SEM image of the micro-region in (f); (d) and (h) Mn and Fe weight ratios of selected spots.

is complicated and the grain sizes of Mn-containing and Fe-containing minerals are very fine. Both in Chinese and South African Fe-Mn ores, there exists one kind of phase with high Fe and Mn contents. In this work, this kind of phase is called as Fe-Mn composite oxide. This is maybe the isomorphism substitution of Fe and Mn elements in the lattice of hematite and pyrolusite during the long-term weathering and leaching under complex geochemical environment or there does still

exist the potential for fine intergrowths of Fe oxide and Mn oxide minerals (Baioumy et al., 2013; Granina et al., 2010). Moreover, this kind of Fe-Mn composite oxides is very fine with grain size of 5–10 μ m. Therefore, it is extremely hard to completely separate Mn and Fe from the Fe-Mn ores only by physical beneficiation methods.

90–95% of the manganese products is utilized in the Iron & Steel industry in the form of Mn alloys. The demand for manganese ores is

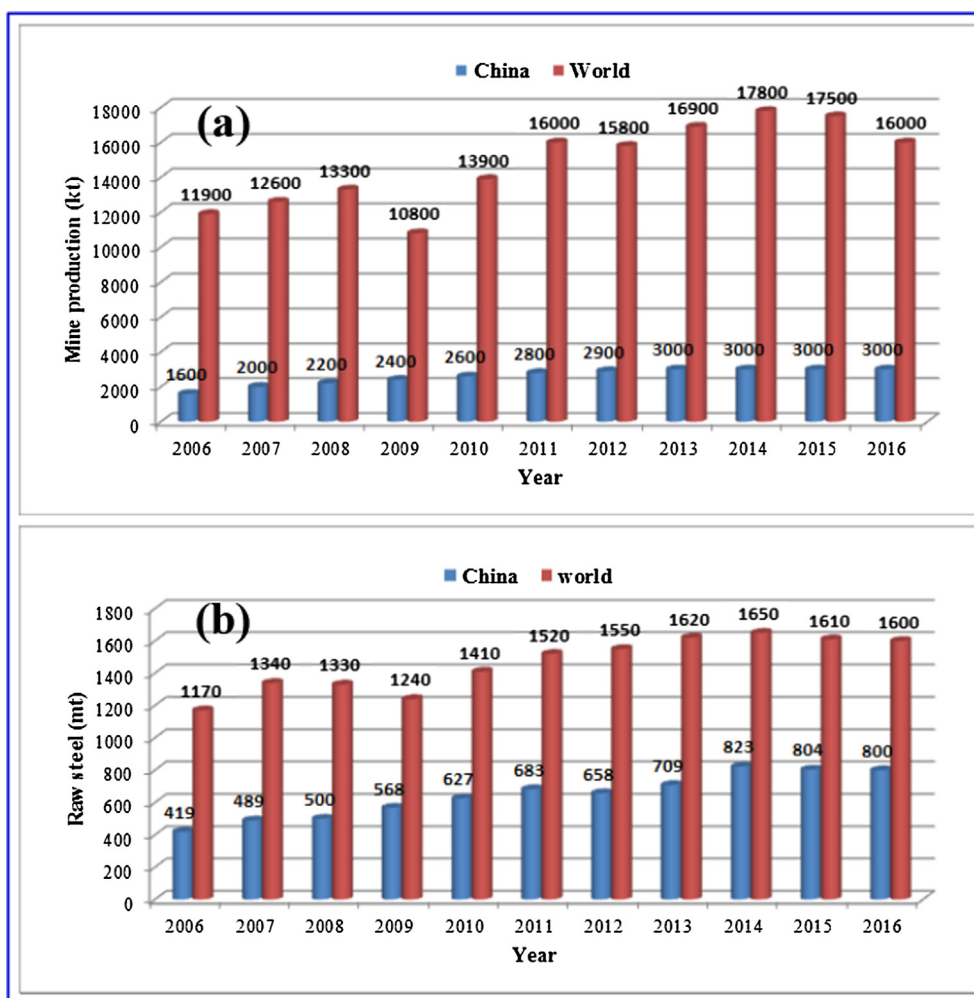


Fig. 3. (a) Manganese mine production, and (b) the crude steel production of the world and China (USGS; IMnI; China Manganese Industry).

consistent with crude steel production. Fig. 3 displays the manganese mine production and crude steel production of the world and China. Global crude steel production remained stable at around 1.6 billion tons in 2016, after reaching a high record of 1.65 billion tons in 2014. China produced around 800 million tons (Mts) crude steel in 2016, down by only 10 Mts from 2015, in spite of the ambitious plan of China's government to reduce steel production to 100–150 Mts per year by 2020. According to Fig. 3(a), global and Chinese Mn ore production has been relatively stable from 2011 to 2016 but with a peak in world production in 2014. Due to the reduction of steel output, the manganese mine production declined from 13.3 Mts in 2008 to 10.8 Mts in 2009, but then it increased rapidly to 13.9 Mts in 2010. Presently, the global manganese mine production is forecast to be 14–16 Mts per year. In addition, the Chinese manganese mine production kept stable at 3 Mts per annum from 2013 to 2016, accounting for 16.9–18.8% of the global mine production.

The main manganese alloy products include silicomanganese (SiMn), high carbon ferromanganese (HCFMn), low and middle ferromanganese (RefFeMn). Fig. 4 shows the distribution proportion of different Mn alloys produced globally from 2008 to 2016 and the total Mn alloys output of the world and China. Global SiMn production increased from 8.4 Mts to 12.5 Mts from 2008 to 2016, while HCMnFe and RefFeMn production remained relatively stable. In 2016, SiMn, HCMnFe and RefFeMn accounted for 72.7%, 19.8% and 7.5% of global production, respectively. The world output of Mn alloys totaled 17.2 Mts in 2016, which represented a 13% increase compared with the output in 2015 but still lower than the record of 18.7 Mts in 2014.

Chinese Mn alloys production was 9.5 Mts in 2016, and increased by 28% from 2015 but still was lower than the record of 10.3 Mts of 2014.

China manganese mine production only accounted for 16.9–18.8% of the global mine production while Chinese smelters produced 55% of the global Mn alloys in 2016. Hence, to moderate the overdependence on import, it's imperative to exploit and utilize the abundant Chinese Fe-Mn ores to ensure the sustainable development of manganese industry (Yan and Gao, 2009; Liu et al., 2015; Zhu, 2016).

3. Processes for extraction and separation of Mn and Fe

3.1. Physical separation

Physical separation is the first step for the separation of Mn and Fe from Fe-Mn ores. At present, common physical separation methods for Fe-Mn ores include washing, gravity, magnetic, flotation, etc. (Wang et al., 2008; Ding et al., 2013).

Washing is the initial procedure to remove clay minerals from Fe-Mn ores and provides preconditions for the smooth conduct of subsequent processing operations (Ding et al., 2013; Tripathy et al., 2015). Gravity separation is based on the density difference of different minerals in the Fe-Mn ores. The densities of the pyrolusite, psilomelane, rhodochrosite and braunite minerals are 4.7–4.8 g/cm³, 4.4–4.7 g/cm³, 3.6–3.7 g/cm³ and 4.7–5.0 g/cm³, respectively. The densities of the hematite, magnetite, siderite and limonite are 4.9–5.3 g/cm³, 4.9–5.2 g/cm³, 3.7–4.0 g/cm³ and 4.0–4.3 g/cm³, respectively (Zhou 2006; Wang et al., 2008). The densities of manganese minerals and iron

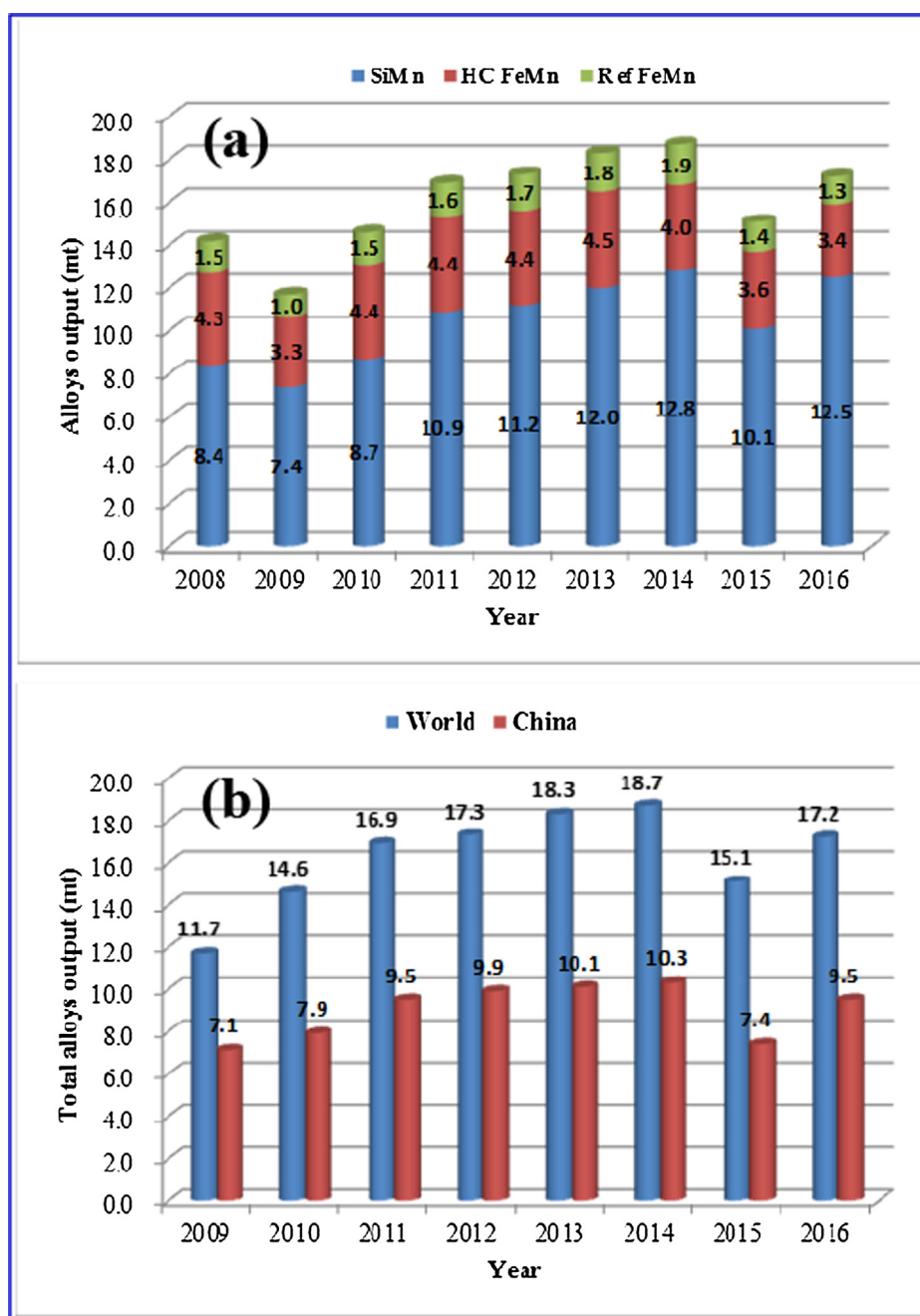


Fig. 4. (a) Distribution proportion of different Mn alloys output of the world, and (b) the total Mn alloys output of the world and China (USGS; IMnI; China Manganese Industry).

minerals are therefore almost at the same level, so separation of Fe and Mn minerals by gravity separation is not good. At the end of the 20th century, some researchers applied the flotation method to separate Mn from Fe-Mn ores, and the flotation behaviors of manganese ores and iron ores indicated that the slurry system was very complex due to the dissolution of Mn^{2+} , Mn^{4+} , Fe^{2+} and Fe^{3+} ions. The redox and hydration of the ions on mineral surfaces deteriorated the flotation process (Wang, 1986; Tang et al., 1987; Wang and Liu, 1993). As reported, only the Zunyi Iron Alloy factory in China built a flotation plant to recover manganese ores using sodium petroleum sulfonate and oxidized paraffin soap (10:1) as mixed collectors (Ding et al., 2013). High intensity magnetic separation was also a common technology to upgrade the Fe-Mn ores considering differences in magnetic susceptibility of different minerals, but the separation effect of Mn and Fe was still poor (Mishra et al., 2009; Mpho et al., 2013; Grieco et al., 2014; Tripathy

et al., 2015).

Table 2 lists the physical separation results of Mn and Fe from typical Fe-Mn ores in China, South Africa and India. The separation results indicate that it's very difficult to effectively separate Mn and Fe, although the gangue minerals can be removed easily and the Mn and Fe grades of the concentrates are improved. In general, due to the complex mineralogy of the Fe-Mn ores (Fig. 2), it's hard to effectively separate Fe from the Fe-Mn ores and achieve manganese concentrates with a high Mn/Fe ratio only using physical beneficiation methods.

3.2. Direct reduction leaching

Manganese sulfate is the most productive manganese compound and is used as a raw material in the production of electrolytic manganese. At the outset, manganese carbonate ores are the main resources utilized

Table 2

Physical separation results of Mn and Fe from the Fe-Mn ores.

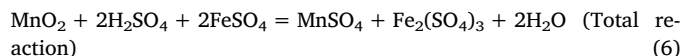
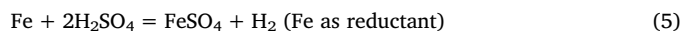
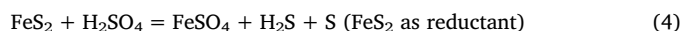
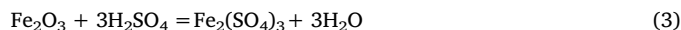
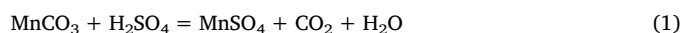
Fe-Mn ores	Separation methods	Feeds (%)			Manganese concentrates				Ref.
		TMn	TFe	Mn/Fe	TMn	TFe	Mn/Fe	RMn	
Zunyi, China	High intensity magnetic	18.38	8.78	2.09	25.23	8.11	3.11	89.70	Zhang (1992)
Zunyi, China	High intensity magnetic – Flotation – Gravity	18.38	8.78	2.09	30.60	5.99	5.10	77–79	
Yunnan, China	Shaking table	19.13	22.23	0.86	29.04	23.20	1.25	35.60	Shi et al. (2007)
Yunnan, China	Jigging	18.93	22.85	0.83	29.55	22.13	1.34	35.10	
Yunnan, China	High intensity magnetic	28.18	24.20	1.16	29.15	22.42	0.73	84.80	
Sichuan, China	High intensity magnetic	19.11	15.50	1.23	23.37	17.88	1.31	81.90	Yu and Zhou (2014)
Hunan, China	High intensity magnetic	10.39	2.93 ^a	3.55	22.75	–	–	89.90	Y. Wu et al. (2015)
Northern Cape, South Africa	High intensity magnetic	29.80	19.22	1.55	29.72	21.18	1.40	59.90	Mpho et al. (2013)
India	High intensity magnetic	32.42	14.13	2.29	42.00	7.63	5.50	47.00–49.00	Singh et al. (2011)
India	Two-stage high intensity magnetic	22.40	14.50	1.54	45.80	14.77	3.10	44.70	Tripathy et al. (2015)
Orissa, India	High intensity magnetic	25.96	2.98 ^b	8.71	42.70	–	–	51.20	Mishra et al. (2009)
Orissa, India	Shaking Table	26.24	2.96 ^c	8.86	42.05	4.34	9.69	40.70	

^a The tested samples are manganese ores with lower Fe content and contain 39.1% SiO₂.^b The tested samples are manganese ores with lower Fe content and contain 32.95% SiO₂.^c The tested samples are manganese ores with lower Fe content and contain 32.50% SiO₂.

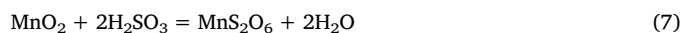
for the extraction of Mn in the electrolytic manganese industry. Manganese carbonate can be dissolved in acid directly since the divalent form of Mn²⁺ is soluble (Reaction (1)) (Momade, 1996; Y.C. Liu et al., 2014; Lu et al., 2016). As high-grade Mn carbonate ores are exhausted gradually, much attention has been paid to recovery of manganese from the Mn oxide ores. In the oxide type manganese ore, the manganese is often present in the insoluble form of Mn⁴⁺ and iron often exists in the state of Fe³⁺. In order to obtain soluble Mn²⁺, Mn⁴⁺ must be converted into Mn²⁺, therefore, reductants are applied to reduce Mn⁴⁺ into Mn²⁺ during the acid leaching process (Named as direct reduction leaching).

A common problem in the leaching process is the co-leaching of iron and manganese (Reactions (2) and (3)). Thus, the downstream purification procedure for the MnSO₄ solution becomes very complicated. During the direct reduction leaching of the Fe-Mn ores, Mn⁴⁺ should be reduced to Mn²⁺ completely while the Fe₂O₃ should not be reduced as much as possible. Hence, the selection of the reductants and leaching agents are very important for the separation of Fe and Mn using this method. Summarizing the current researches on the direct reduction leaching processes for manganese ores, the reductants include inorganic reductants (pyrite, SO₂, Fe/FeSO₄, H₂O₂, CaS, etc.) (Nayak et al., 1999; Vračar and Cerović, 2000; Naik et al., 2000, 2003; Senanayake, 2004; El-Hazek et al., 2006; Bafghi et al., 2008; Nayl et al., 2011; Li et al., 2015; Deng et al., 2017) and organic reductants (oxalic acid, methanol, lactose, glucose, and biomass, etc.) (Momade and Momade, 1999a, 1999b; Sahoo et al., 2001; Ismail et al., 2004; Su et al., 2008; Tang et al., 2014); the leaching agents cover hydrochloric acid, sulfuric acid and nitric acid (Pagnanelli et al., 2004; El-Hazek et al., 2006; Nayl et al., 2011).

Table 3 shows the leaching results of the Fe-Mn ores by different reductants and acids. On the whole, the reduction of Mn⁴⁺ to Mn²⁺ is effective using all the reductants used in an acid system and the leaching efficiency of Mn is over 90%. Nevertheless, the leaching efficiency of Fe depends on the reductant type and acid concentration. Leaching efficiency of Fe increases with the acid concentration, which is as high as 84% in a 4 mol/L H₂SO₄ solution (Nayl et al., 2011). Comparatively, the leaching efficiency of Fe is lower than 10% when the Fe-Mn ores are reduced by oxalic acid, gaseous SO₂ and methanol. As for the Fe-Mn ores, pyrite, metallic Fe and FeSO₄ are not recommended as the reductants since plenty of iron ions are introduced into the leaching solution from the reductants (Nayak et al., 1999; Vračar and Cerović, 2000; Bafghi et al., 2008), and the main Reactions (4), (5) and (6) are listed as the following. It is both complex and capital intensive to remove the iron ions in the following purification procedure.



Although the direct reduction leaching of manganese ores achieves about 90% Mn leaching efficiency and 80% Fe leaching efficiency, this process can be used for leaching manganese ores with lower iron content (Pagnanelli et al., 2004; Tian et al., 2010; Azizi et al., 2012; Y.C. Liu et al., 2014; Tang et al., 2014). For the high-grade manganese ores with lower iron content, even though all of the iron elements are dissolved into the solution, the iron ion concentration of the solution is still low. On the contrary, as for the Fe-Mn ores with high iron content, the iron ions concentration of the solution is high even if the iron leaching efficiency is only 20–30%. The leaching of Fe-Mn ores in the presence of oxalic acid and methanol presents better selective leaching effect of manganese and iron, but it's uneconomic owing to the high price of the organic reductants. In addition, the leaching temperature is relatively higher (85–160 °C) resulting in the energy-intensive consumption and strict requirement of leaching devices. Gaseous SO₂ also has a good selectivity for the reduction leaching of Fe-Mn ores, whereas it's inevitable that MnS₂O₆ generates according to the following Reaction (7), deteriorating the quality of the MnSO₄ product (Chen, 2010; Yang and Zhang, 2012).



In order to reduce the MnS₂O₆ content in the MnSO₄ products, further purification is conducted by heating the solid crystals of MnS₂O₆-MnSO₄ at 140–190 °C for 120 min (Reaction (8)) (Lei et al., 1989) or leaching the MnS₂O₆-MnSO₄ solution with the addition of pyrolusite (MnO₂) (Reaction (9)) (Chen, 2010). As for the heat treatment process, the treatment should be controlled in the water vapor atmosphere and the dehydration of MnSO₄·H₂O during the heat treatment in turn affects the product quality. As for the leaching method, about 91% MnS₂O₆ transforms into MnSO₄ when 12.2 g pyrolusite and 20 mL H₂SO₄ are added into a 200 mL solution (104 g/L MnSO₄ and 51.8 g/L MnS₂O₆) with leaching conducted at 100 °C for 180 min with an agitation speed of 500 rpm.

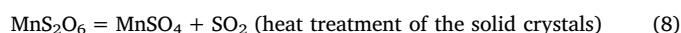


Table 3
Direct reduction leaching results of different Fe-Mn ores and the main reactions.

Chemical compositions (wt.%)				Reduction leaching conditions		Leaching efficiency (wt.%)		Main reactions		Ref.
Fe	Mn	SiO ₂	T (°C)	Time (min)	Reductant	Leaching medium	Fe	Mn		
16.5	32.5	–	29	15	3.5 vol% SO ₂	~0.08 mol/L H ₂ SO ₄ , L/S = 20:1	4.0	96.0	SO ₂ + H ₂ O = H ₂ SO ₃ MnO ₂ + H ₂ SO ₃ = MnSO ₄ + H ₂ O MnO ₂ + 2H ₂ SO ₃ = MnS ₂ O ₆ + 2H ₂ O	Naik et al. (2000, 2003) Deng et al. (2017)
19.2	20.7	–	35	120	5 vol% SO ₂ + 1% O ₂	Water, L/S = 4:1	2.6	93.1		
28.4	24.7	–	85	105	30.6 g/L oxalic acid	0.534 mol/L H ₂ SO ₄	8.7	98.4	MnO ₂ + H ₂ C ₂ O ₄ + 2H ⁺ = Mn ²⁺ + 2CO ₂ + 2H ₂ O	Sahoo et al. (2001)
36.4	16.0	4.72	90	90	50% Sawdust	30% H ₂ SO ₄	35.0–40.0	92.5	24MnO ₂ + C ₁₂ H ₂₂ O ₁₁ + 4H ₂ SO ₄ = 24MnSO ₄ + 12CO ₂ + 35H ₂ O 12MnO ₂ + C ₆ H ₁₂ O ₆ + 12H ₂ SO ₄ = 12MnSO ₄ + 6CO ₂ + 18H ₂ O; Fe ₂ O ₃ + 3H ₂ SO ₄ = Fe ₂ (SO ₄) ₃ + 3H ₂ O	Ismail et al. (2004)
36.4	16.0	4.72	90	120	75% Lactose	20% H ₂ SO ₄	25.0–30.0	90.5	24MnO ₂ + C ₁₂ H ₂₂ O ₁₁ + 4H ₂ SO ₄ = 24MnSO ₄ + 12CO ₂ + 35H ₂ O 12MnO ₂ + C ₆ H ₁₂ O ₆ + 12H ₂ SO ₄ = 12MnSO ₄ + 6CO ₂ + 18H ₂ O; Fe ₂ O ₃ + 3H ₂ SO ₄ = Fe ₂ (SO ₄) ₃ + 3H ₂ O	Ismail et al. (2004)
7.4–8.6	25.9–32.0	–	160	120	40 vol% methanol	0.3 mol/L H ₂ SO ₄	0.4–3.5	98.0	MnO ₂ + CH ₃ OH + 2H ⁺ = Mn ²⁺ + HCHO + 2H ₂ O	Momade and Momade (1999a, 1999b) El-Hazek et al. (2006)
14.1	8.52	16.0	60	60–95	0.4 mol/L H ₂ O ₂	2 mol/L HCl	14.0	97.0	MnO ₂ + H ₂ O ₂ + 2H ⁺ = Mn ²⁺ + 2H ₂ O + O ₂	Su et al. (2008)
12.4	22.5	24.2	90	120	60.0 g/L Cane molasses	1.9 mol/L H ₂ SO ₄	32.4	97.0	24MnO ₂ + C ₁₂ H ₂₂ O ₁₁ + 4H ₂ SO ₄ = 24MnSO ₄ + 12CO ₂ + 35H ₂ O 12MnO ₂ + C ₆ H ₁₂ O ₆ + 12H ₂ SO ₄ = 12MnSO ₄ + 6CO ₂ + 18H ₂ O	Nayl et al. (2011)
16.1	39.7	6.8	40	90	0.8 mol/L H ₂ O ₂	4 mol/L H ₂ SO ₄ , L/S = 5:1	84.0	92.0	MnO ₂ + H ₂ O ₂ + 2H ⁺ = Mn ²⁺ + 2H ₂ O + O ₂ ; Fe ₂ O ₃ + 3H ₂ SO ₄ = Fe ₂ (SO ₄) ₃ + 3H ₂ O	Li et al. (2015)
11.5	20.6	37.6	95	120	25% CaS ^a	1.5 mol/L H ₂ SO ₄ , 80 °C, 5 min	19.2	96.5	2CaS + 3MnO ₂ + 2H ₂ O = Mn ₃ O ₄ + 2Ca(OH) ₂ + 2S 2Mn ₃ O ₄ + 6H ₂ SO ₄ = 6MnSO ₄ + 6H ₂ O + O ₂	

^a Two-stage leaching, the first is the Fe-Mn ore and CaS leaching in water at 95 °C for 120 min; the second is the residue after filtration leaching in 1.5 mol/L H₂SO₄ at 80 °C for 5 min.

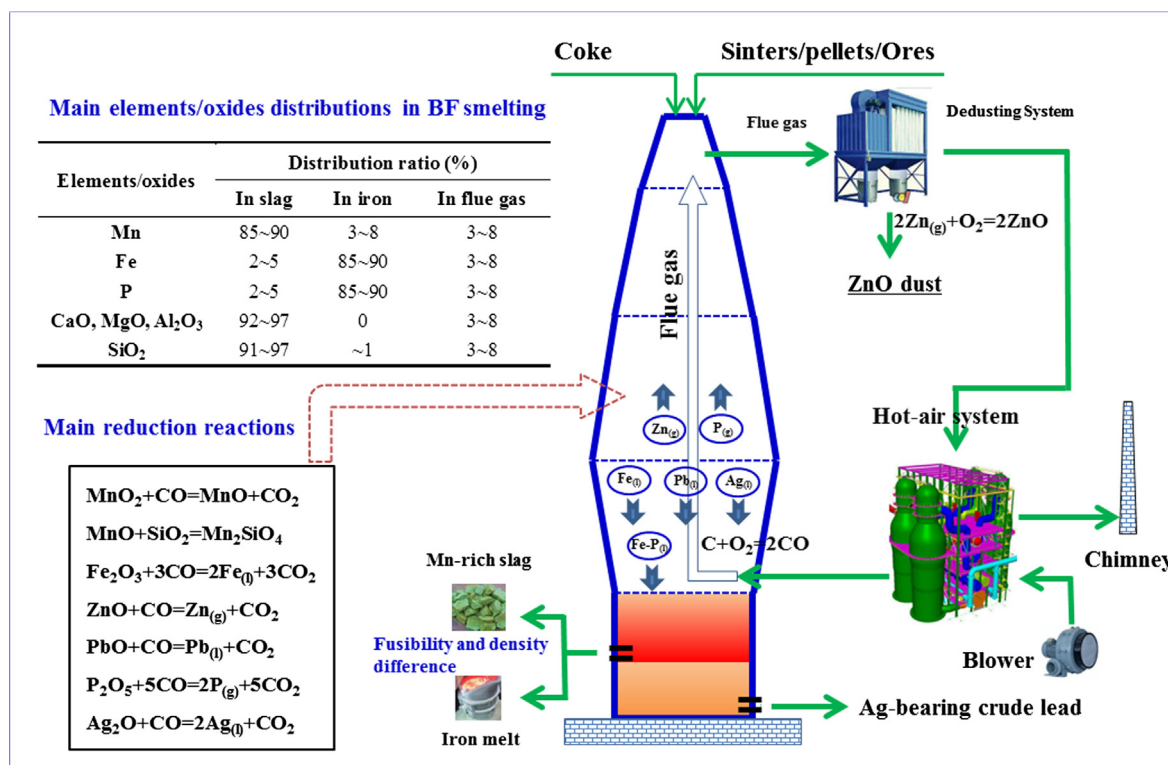
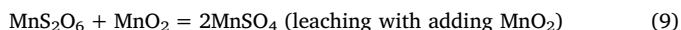


Fig. 5. Flow sheet of Mn rich slag smelting process in blast furnace (Inset table is the main elements/oxides distributions in BF smelting) (Tan et al., 2004; Mei et al., 2011).



3.3. Mn-rich slag smelting

High temperature reduction followed by smelting separation (also named as Mn-rich slag smelting) is the main and most mature pyrometallurgical process for the separation of Mn and Fe from the Fe-Mn ores (Tan et al., 2004; Mei et al., 2011). The fusibility difference between the manganese silicates slag and molten iron or iron carbides is the key point for the separation of Mn and Fe. The high temperature reduction and smelting separation procedures are conducted in one furnace, such as the blast furnace (BF) or electric arc furnace (EAF).

In China, decades of production practice has testified this process is suitable for utilizing the Fe-Mn ores with high contents of Si, Zn, Pb, P and Ag. The flow sheet of Mn-rich slag smelting in the BF is shown in Fig. 5. Raw materials composed of lump Fe-Mn ores, sinters, pellets, coke and silica flux are used (Tan et al., 2004; X.D. Wu et al., 2015; Ming et al., 2016; Zhang et al., 2016a). The basicity of the mixed burden is natural basicity below 0.4, and the operation temperature range in the BF is 1250–1350 °C. Within the reduction zone, most of the zinc oxides and partial phosphorus oxides are reduced to gaseous Zn and P, which have higher saturated vapor pressures and evaporate into the dust fraction with the upward gas flow. MnO₂ is only reduced to MnO and then combines with SiO₂ to form Mn-rich slag (MnO-SiO₂). Fe₂O₃, PbO and Ag₂O are reduced to their respective metallic states. The molten metals have different densities and so form different layers. After smelting separation, the iron melt and Mn-rich slag are achieved. The produced Mn-rich slag has a 30–40% Mn grade, Mn/Fe mass ratio of 10–40 and P/Mn mass ratio < 0.002, which has been proved to be high-quality raw materials for Mn alloy production (Eissa et al., 2004; El-Faramawy et al., 2004a, 2004b; Ahmed et al., 2014, 2015). Table 4 shows the results of the Mn-rich slag smelting for Fe-Mn ores in a 13 m³ BF in Citic Dameng Ming Industries Limited Company, China. The yield of the Mn-rich slag is 73–77% and the recovery of Mn is as high as

85–90%. The coke consumption is 0.7–0.8 t/t_{slag}. The Mn-rich slag smelting method is characterized as large production and high efficiency, while it has defects in the energy-intensive consumption and heavy pollution.

As estimated by Yan (2008), there are about 140 enterprises producing Mn-rich slag in China, and the total output is about 1500–2800 kilotons per year. The main smelting furnaces for production of Mn-rich slag are small blast furnaces with a volume of 9–60 m³ (Wu, 2002; Yao, 2005). Due to the good adaptability for raw materials, low capital cost and good manoeuvrability, the application of the Mn-rich slag smelting process is of great importance for the effective utilization of the abundant Fe-Mn ores presented in China.

3.4. Carbothermic reduction roasting followed by magnetic separation (or acid leaching)

Carbothermic reduction roasting followed by magnetic separation (or acid leaching) processes have been regarded as effective routes to realize the separation of Mn and Fe via enlarging the magnetic property and solubility differences among Mn, Fe and gangue constituents. During these processes, the carbothermic reduction roasting is the critical procedure for separating Mn and Fe. In the past decades, numerous studies were carried out to illustrate the reduction behaviors of iron oxides and manganese oxides by various reductants, such as solid reductive agents (graphite, coke, biomass, coal, etc.) (Hansen et al., 1993; Swamy et al., 1998; Halim, 2007; Cheng et al., 2009; Zhang et al., 2012a; Halim et al., 2013; Duan, 2013; El-Hussiny et al., 2015; Gao et al., 2018), and gaseous reductive agents (CO, H₂, CH₄, etc.) (Barner and Mantell, 1968; El-Geassy et al., 2000, 2007, 2008; Anacleto et al., 2004; Ostrovski and Zhang, 2006; Kononov et al., 2008; Gao et al., 2012a, 2012b; Liu et al., 2017a, 2018a, 2018b).

The gas-phase equilibrium diagram of MnO₂ and Fe₂O₃ under various CO-CO₂ atmospheres is plotted in Fig. 6. The MnO₂ and Fe₂O₃ can be reduced stepwise in the orders of MnO₂ → Mn₂O₃ → Mn₃O₄ → MnO

Table 4
Results of the Mn rich slag smelting in a 13 m³ BF in Citic Dameng Ming Industries Limited Company, China (X.D. Wu et al., 2015; Ming et al., 2016).

Batch No.	Chemical compositions of the burdens (wt.%)							Chemical compositions of the Mn slag (wt.%)							Yield of Mn rich slag	Recovery of Mn	Coke consumption (t/t _{slag})				
	Mn	Fe	P	CaO	MgO	Al ₂ O ₃	SiO ₂	Mn/Fe	(CaO + MgO)/SiO ₂	Mn	Fe	P	CaO	MgO				Al ₂ O ₃	SiO ₂	Mn/Fe	P
1	18.05	32.47	0.41	1.26	0.54	4.58	13.91	0.56	0.13	39.46	2.33	0.01	1.82	0.40	11.21	28.62	16.94	0.76	73.00–77.00	95.90	0.7–0.8
2	20.07	30.90	0.39	1.26	0.72	4.36	13.19	0.65	0.15	41.29	1.11	0.01	1.40	0.60	10.96	26.33	37.20	0.76	73.00–77.00	92.37	0.7–0.8
3	22.10	29.32	0.37	1.26	0.91	4.13	12.47	0.75	0.17	42.08	1.55	0.01	2.38	0.71	11.47	30.99	27.15	0.78	73.00–77.00	87.30	0.7–0.8

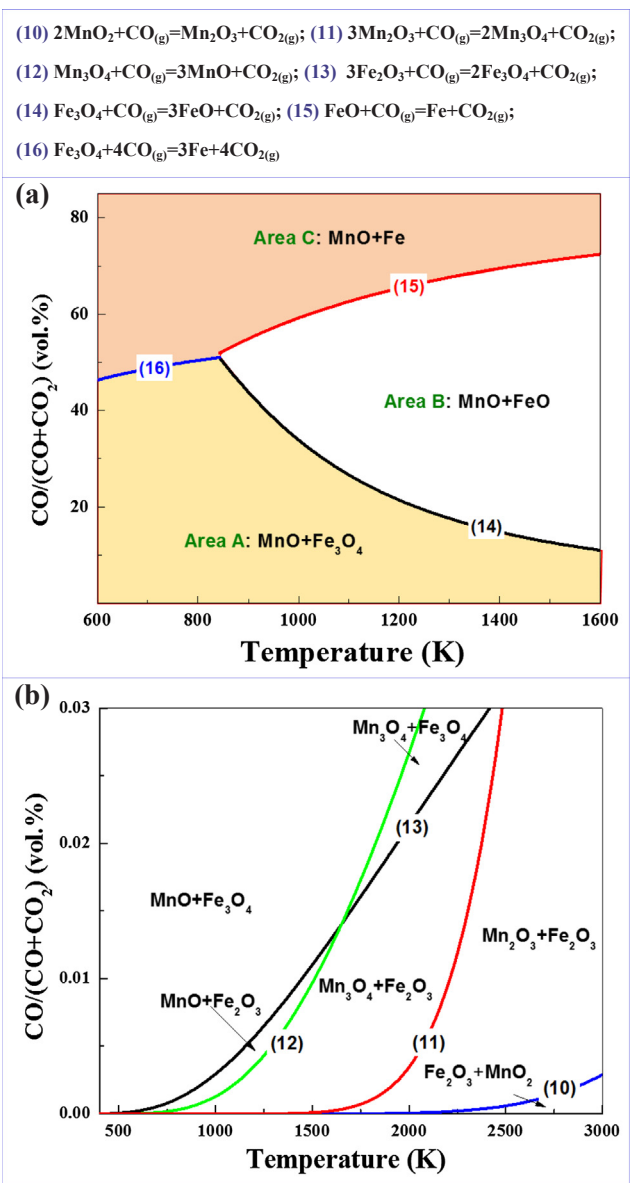


Fig. 6. Gas-phase equilibrium diagram of MnO₂-Fe₂O₃-CO-CO₂ system.

and Fe₂O₃ → Fe₃O₄ → FeO → Fe (Fe₃C), respectively. The reduction temperatures and CO contents required for the stepwise reduction of Fe₂O₃ and MnO₂ are different. Fe₂O₃ and Fe₃O₄ have weak solubility in a dilute acid solution. However, in comparison with Fe₂O₃ and Fe₃O₄, FeO and Fe are more easily dissolved in the dilute acid solution. Fe₃O₄ and metallic Fe have strong magnetism compared to MnO₂ and Mn₃O₄. Based on the reduction state of MnO₂ and Fe₂O₃ and the properties of the reduced products, the carbothermic reduction roasting followed by separation process are divided into a magnetic/direct reduction roasting step followed by a magnetic separation process (Gao et al., 2012a, 2012b, 2018; Zhang et al., 2017a), and magnetic reduction followed by acid leaching (Zhang et al., 2015). Therefore, the optimal magnetic reduction roasting conditions should be restricted to Area A in Fig. 6, where the MnO₂ is only reduced to acid-soluble and non-magnetic MnO while Fe₂O₃ is only reduced to Fe₃O₄. However, the direct reduction roasting conditions should be controlled in Area C, where is the coexisting area of MnO and metallic Fe.

3.4.1. Magnetic separation

The magnetic/direct reduction roasting procedure for Fe-Mn ores is conducted in a rotary kiln. Fig. 7 shows the flow sheet of the magnetic/

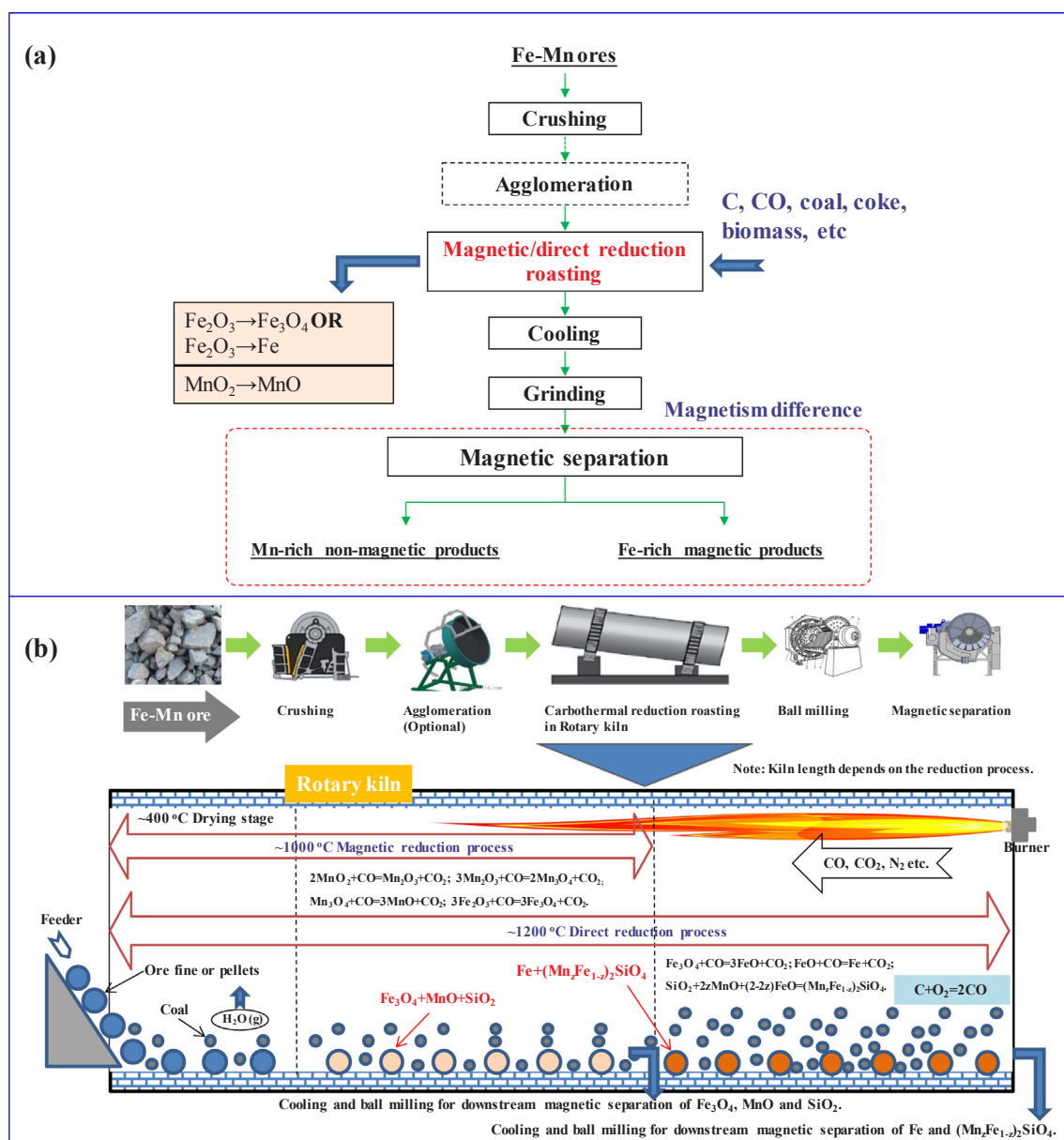


Fig. 7. (a) Flow sheet of magnetic/direct reduction roasting-magnetic separation process, (b) Schematic diagram of the industrial separation of Mn and Fe from Fe-Mn ores and the reduction reactions of Fe-Mn ores inside the rotary kiln.

direct reduction roasting-magnetic separation process, the schematic diagram of the industrial separation of Mn and Fe from Fe-Mn ores and the main reduction reactions inside the rotary kiln. The magnetic reduction roasting procedure is usually conducted at 400–1000 °C under a weakly reducing atmosphere (low consumption of reductants). The direct reduction roasting procedure is conducted under strongly reducing atmosphere (high consumption of reductants) at an elevated temperatures range of 900–1200 °C. The magnetic Fe-rich products can be used as raw materials for iron-making and steel-making, and the non-magnetic Mn-rich products can be used as high-quality burdens for Mn alloys production or the production of electrolytic manganese.

Table 5 shows the results summary of Mn and Fe separation from Fe-Mn ores by magnetic/direct reduction roasting followed by a magnetic separation process. As shown in Table 5, Mn and Fe is somewhat separated by the magnetic reduction process, and the Mn grade and Mn/Fe mass ratio of the Mn-rich products increase dramatically compared with the raw Fe-Mn ores. However, the main problem is the magnetic Fe-rich products with a high Mn content. In the research

conducted by Gao et al. (2012a, 2012b), the Mn content of the magnetic products was as high as 36.1%. As for the direct reduction roasting without additives, the Mn/Fe mass ratio in the non-magnetic Mn-rich products is enhanced to 10–30, nevertheless the Mn content in the magnetic products is still as high as 18–27 wt% (Zhao, 2012; Cai, 2013). The final Fe-rich products can't be directly used as raw materials for steel-making in the electric furnace (EF).

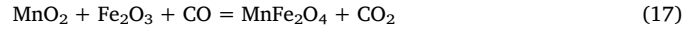
Considerable studies about the magnetic/direct reduction roasting of Fe-Mn ores has been conducted in Central South University (CSU) to enhance the separation of Mn and Fe (Zhao, 2012; Cai, 2013; Zhang et al., 2017a; Yi et al., 2017). Generally, the reasons for the incomplete separation of Mn and Fe include that the magnetic and non-magnetic products are not liberated during the milling process or the particle size of the magnetic feed substance is too small after grinding (Wang et al., 2008; Tripathy et al., 2015; Yao et al., 2015). Nevertheless, our group has developed some new understandings about the essence of the magnetic reduction roasting process of Fe-Mn ores (Liu et al., 2017b, 2018b, 2018c). Fig. 8 presents the schematic diagram of the phase

Table 5
Results summary of Mn and Fe separation from Fe-Mn ores by magnetic/direct reduction roasting followed by magnetic separation.

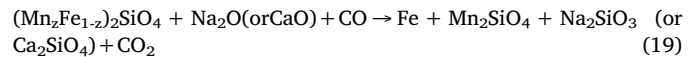
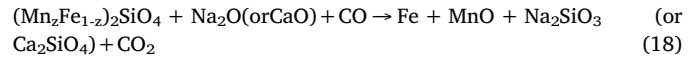
Feed (wt.%)		Reduction conditions				Reduction process	Magnetic product				Non-Magnetic product				Ref.					
		Mn	Fe	SiO ₂	Mn/Fe		T (°C)	Time (min)	Reductants	Additives	Yield	TFe	TMn	RFe		Yield	TFe	TMn	RMn	Mn/Fe
32.24	14.13	20.76	2.29	650	30	15% coal			None	Magnetic	–	–	–	–	7.67	46.00	–	6.00	Singh et al. (2011)	
27.71	30.90	4.14	0.90	800	480	10%CO + 90%CO ₂			None	Magnetic	62.91	51.10	13.20	97.18	2.52	58.44	72.31	23.19	Mpho et al. (2013)	
36.6	12.7	11.7	2.88	600	25	10%CO + 90%CO ₂			None	Magnetic	4.90	32.80	36.10	~70.00	95.10	3.40	45.20	~96.00	13.29	Gao et al. (2012a, 2012b)
44.12 ^a	21.14	0	2.08	1100	100	Excessive anthracite			None	Direct	41.49	70.09	23.18	94.38	58.51	2.96	70.83	81.15	23.93	Zhao (2012)
44.12 ^a	21.14	0	2.08	1100	100	Excessive anthracite			1% KCl	Direct	37.59	78.59	15.38	94.66	62.41	2.67	71.67	88.55	26.84	
44.12 ^a	21.14	0	2.08	1100	100	Excessive anthracite			1% BaCl ₂	Direct	35.06	80.63	13.75	94.28	64.94	2.64	71.62	90.60	27.13	
44.12 ^a	21.14	0	2.08	1100	100	Excessive anthracite			1% Na ₂ B ₄ O ₇ ·10H ₂ O	Direct	34.72	83.46	12.61	95.25	65.28	2.21	71.75	91.44	32.46	
39.05	21.28	4.79	1.84	1100	100	Excessive anthracite			None	Direct	41.96	~68.00	~22.00	~96.00	58.04	2.04	59.70	79.29	29.26	
39.05	21.28	4.79	1.84	1100	100	Excessive anthracite			1% Na ₂ B ₄ O ₇ ·10H ₂ O	Direct	38.62	~75.00	~18.00	~95.00	61.38	2.27	61.37	84.53	27.04	
39.05	21.28	4.79	1.84	1100	100	Excessive anthracite			4% Na ₂ B ₄ O ₇ ·10H ₂ O	Direct	32.79	~83.00	~13.00	~94.00	67.21	2.27	58.55	91.45	25.79	
34.73	23.48	6.75	1.48	1100	100	300% lignite			None	Direct	53.37	59.42	20.73	89.32	46.63	5.91	48.88	71.94	8.27	Zhang et al. (2017a)
34.73	23.48	6.75	1.48	1100	100	300% lignite			5 wt% Na ₂ CO ₃	Direct	44.75	74.30	14.65	93.67	55.25	3.52	54.31	84.06	15.43	
34.73	23.48	6.75	1.48	1100	100	300% lignite			5 wt% Na ₂ S	Direct	40.60	81.94	11.24	87.76	59.4	3.57	46.88	87.23	13.13	
34.73	23.48	6.75	1.48	1100	100	300% lignite			5 wt.%Na ₂ SO ₄	Direct	41.76	79.87	11.08	93.93	58.24	3.36	54.56	84.71	16.24	
34.73	23.48	6.75	1.48	1100	100	300% lignite			5wt.%Na ₂ S ₂ O ₃	Direct	40.94	82.45	10.29	95.05	59.06	3.01	54.72	83.65	18.18	
34.73	23.48	6.75	1.48	1100	100	300% lignite			7wt.%Na ₂ S ₂ O ₃	Direct	~40.00	86.39	6.45	97.04	~60.00	1.93	54.84	88.01	28.41	

^a In this test, the raw material is the mixture of pure MnO₂ and Fe₂O₃.

transformations of the Fe₂O₃-MnO₂-SiO₂ system during the magnetic/direct reduction roasting processes. It's demonstrated that MnFe₂O₄ can be generated from MnO₂ and Fe₂O₃ reduced under 10 vol% CO over 800 °C according to Reaction (17). The MnFe₂O₄ phase has a strong magnetism. Meanwhile, the MnFe₂O₄ and Fe₃O₄ as well as silicates and SiO₂ are tightly integrated as a whole, resulting in the inferior separation of Mn and Fe. To avoid the formation of MnFe₂O₄, low temperatures (below 800 °C) and long reduction times are recommended for the magnetic reduction roasting of Fe-Mn ores.



As for the direct reduction roasting process, the inferior separation of Fe and Mn is owing to the formation of hard reductive (Mn_zFe_{1-z})₂SiO₄ phases. However, the separation effect can be improved with adding some additives such as CaO (Tang et al., 1999a, 1999b), chlorine salts (KCl and BaCl₂) (Cai, 2013), and sodium salts (Na₂CO₃, Na₂S, Na₂SO₄, Na₂S₂O₃ and Na₂B₄O₇·10H₂O) (Cai, 2013; Zhang et al., 2017a) during the direct reduction process. Among these additives, the sodium salts are testified as the most effective ones, as shown in Table 5. Fig. 9 shows the SEM images of the Fe-Mn ores (containing 23.48 wt% TFe, 34.73 wt% TMn and 6.75 wt% SiO₂) reduced in the presence and absence of sodium additives under 100 vol% CO at 1100 °C for 100 min. When the Fe-Mn ores are reduced in the absence of sodium additives, the metallic iron particles are dispersed and a lot of (Mn_zFe_{1-z})₂SiO₄ phases are formed. If the sodium additives are added with the Fe-Mn ores during the reduction process, most metallic iron particles are aggregated together to over 50 μm and the proportion of (Mn_zFe_{1-z})₂SiO₄ phases decreases dramatically. The magnetic metallic iron powders with an Fe grade of 88% can meet the requirement of direct reduction iron (DRI) products for steelmaking. In short, the sodium additives have two functions: (1) the structure of the (Mn_zFe_{1-z})₂SiO₄ phase is destroyed because of the reaction between the alkaline additives and SiO₂ in the (Mn_zFe_{1-z})₂SiO₄, which also promotes the reduction of FeO to metallic iron according to the following Reactions (18) and (19); (2) the B, Na and S constituents in the additives facilitate the growth of metallic iron grains to over 50 μm, which creates a favorable substance condition for downstream magnetic separation process (Li et al., 2012, 2013, 2014a, 2014b; Rao et al., 2013; M.D. Liu et al., 2014).



3.4.2. Acid leaching

In comparison with FeO and metallic Fe, Fe₃O₄ is more difficult to dissolve in a dilute H₂SO₄ solution. Under a suitable reduction atmosphere, Fe₂O₃ can only be reduced to Fe₃O₄ and MnO₂ is reduced to MnO. Therefore, it is feasible to extract Mn from the Fe-Mn ores via magnetic reduction roasting followed by acid leaching. During the magnetic reduction process, biomass resources, including, charcoal, cornstalk, etc., are used as reductants for the reduction of Fe-Mn ores (Sharma, 1992; Zhang et al., 2005, 2015; Cheng et al., 2009). Cheng et al. (2009) reported the detailed parameters for the reduction roasting followed by sulfuric acid leaching of Fe-Mn ores. Mn leaching efficiency reaches 90.2% under the optimal conditions: Fe-Mn ores to cornstalk weight ratio of 10:3, roasting temperature of 500 °C, roasting time of 80 min, leaching stirring speed of 400 rpm, H₂SO₄ concentration of 3 mol/L, leaching temperature of 50 °C and leaching time of 40 min.

Bitumite and anthracite are also used as the reductants in the magnetic reduction process (Feng et al., 2009; Duan, 2013; Zhang et al., 2015). Compared with anthracite, bitumite shows a better ability to reduce Fe-Mn ores and its Mn leaching efficiency is higher. It's found that the bitumite contains a higher sulphur content than anthracite

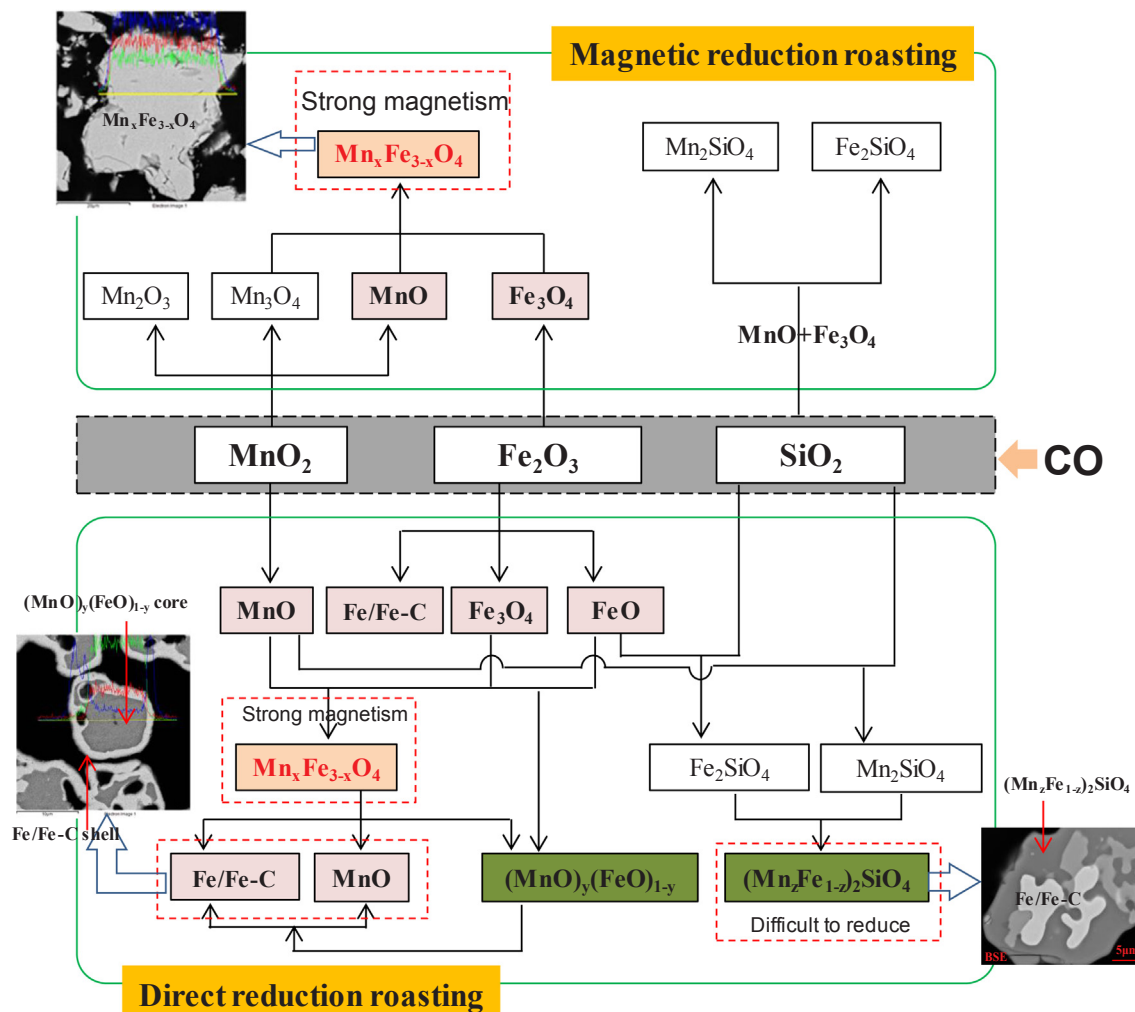


Fig. 8. Schematic diagram of the phase transformation of $\text{Fe}_2\text{O}_3\text{-MnO}_2\text{-SiO}_2$ system during the magnetic/direct reduction roasting process (Liu et al., 2018b, 2018c).

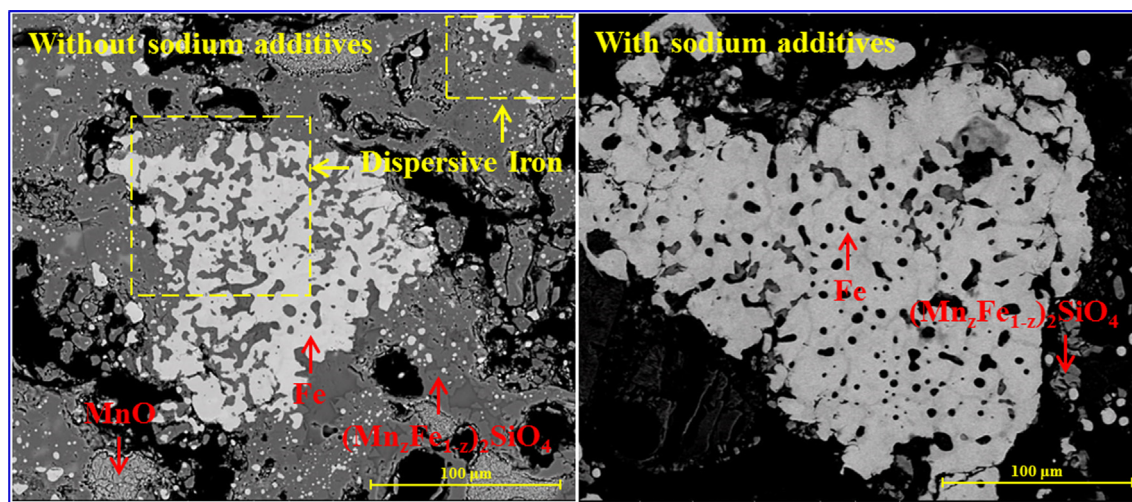


Fig. 9. SEM images of the Fe-Mn ores reduced with and without sodium additives in the direct reduction process (Zhang et al., 2017a).

which is beneficial for the reduction of manganese oxides. Zhang et al. (2015) compared the magnetic reduction roasting and acid leaching behaviors of Fe-Mn ores reduced by black charcoal and anthracite. It's demonstrated that the reductants with good reaction properties were not suitable for the reduction of Fe-Mn ores since the formed Fe_3O_4 was further readily over-reduced to wustite (FeO). Black charcoal showed a

higher Mn leaching efficiency than anthracite due to the relatively bad reaction properties of black charcoal. The Mn leaching efficiency reached 82.4% while the Fe leaching efficiency was maintained below 7% when the Fe-Mn ores were roasted at 650 °C for 40 min with 10% charcoal dosage and the roasted residues were leached in 2.0 mol/L H_2SO_4 at 25 °C for 20 min.

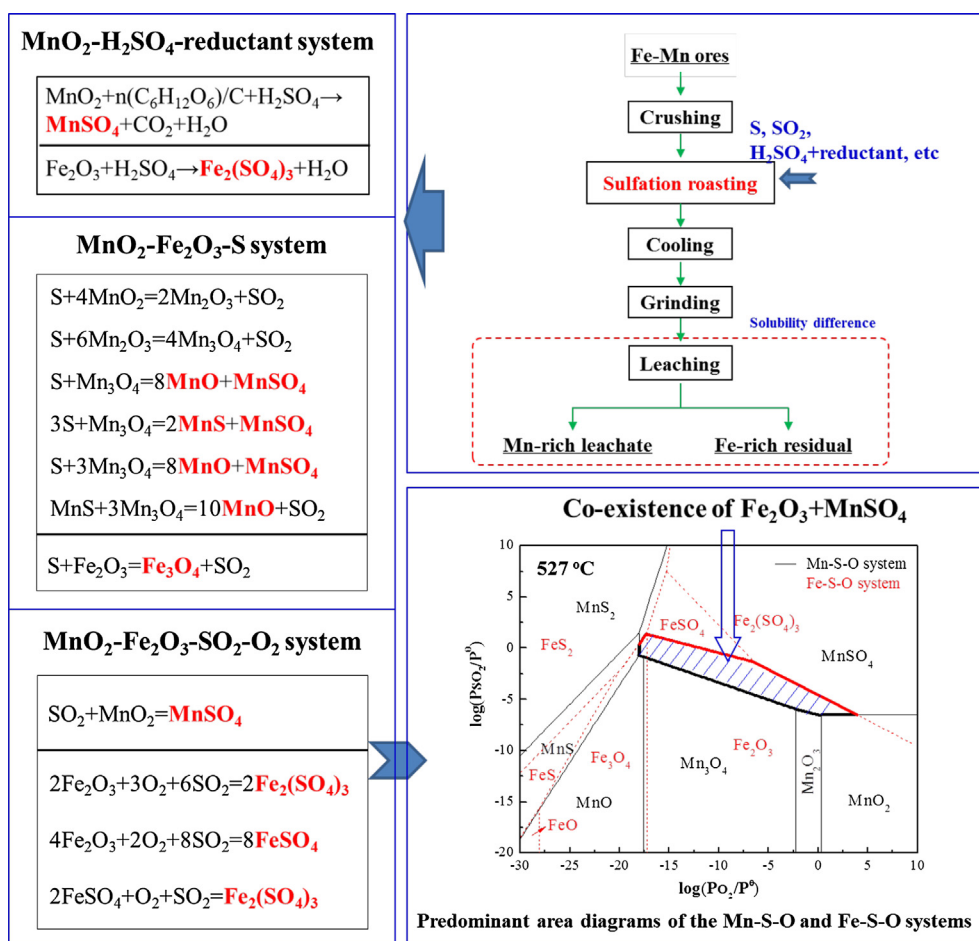


Fig. 10. Flow sheet of the selective sulfation roasting followed by leaching process of Fe-Mn ores and the main reactions of the MnO₂-H₂SO₄-reductant system, MnO₂-Fe₂O₃-S system and MnO₂-Fe₂O₃-SO₂-O₂ system (Bolsaitis and Nagata, 1980; Cai et al., 2012; Ye et al., 2013; Zhang et al., 2013; You et al., 2015, 2017).

The magnetic reduction roasting followed by acid leaching of Fe-Mn ores is usually conducted at a low temperature range of 500–700 °C. At this temperature range, the gangue components (SiO₂, Al₂O₃, etc.) in the Fe-Mn ores seldom react with Fe₃O₄. In order to obtain a product with a higher Fe content from the leaching residue, magnetic separation can be subsequently applied to upgrade and remove the gangue minerals (Feng et al., 2009).

3.5. Sulfation roasting followed by leaching

Sulfation roasting followed by leaching is an alternative process for the extraction of Mn from Fe-Mn ores (Cai et al., 2012; Zhang et al., 2012b, 2013; Jiang et al., 2012a, 2012b, 2016; You et al., 2015, 2017, 2018). In this review, the roasting processes of manganese ores with FeS₂, concentrated H₂SO₄, (NH₄)₂SO₄, sulfur and SO₂ are collectively known as sulfation roasting. Sulfation roasting with FeS₂ and (NH₄)₂SO₄ are rarely used to deal with the Fe-Mn ores. Judging from the extraction of Fe and Mn from a secondary resource (The leaching residue from desulfurization and denitrification of exhaust gas treatment process with manganese ore as absorbent) including pyrolusite and goethite with a low Mn content (30.1%Fe and 4.0%Mn) (Deng et al., 2018), as well as manganese ores with a low Fe content (Yang et al., 2006) by sulfation roasting with (NH₄)₂SO₄, the selective separation of Mn and Fe is ineffective since both Fe and Mn leaching efficiencies are over 90%. Hence, sulfation roasting with (NH₄)₂SO₄ is not recommended to deal with Fe-Mn ores for the selective separation of Mn and Fe.

Fig. 10 describes the flow sheet of the selective sulfation roasting of

Fe-Mn ores and summarizes the main reactions of the MnO₂-H₂SO₄-reductant system, MnO₂-Fe₂O₃-S system and MnO₂-Fe₂O₃-SO₂-O₂ system. The sulfation roasting procedure for Fe-Mn ores is crucial to the selective leaching of Mn and Fe via controlling the selective reduction of MnO₂ and Fe₂O₃. As for the MnO₂-H₂SO₄-reductant system, almost all iron-bearing minerals and manganese-bearing minerals are dissolved into the molten H₂SO₄ at a higher roasting temperature. Hence, the water leaching efficiencies of Fe and Mn approach 100% (Cai et al., 2012). In the MnO₂-Fe₂O₃-S system, MnO₂ is reduced to MnO, MnS and MnSO₄ by sulfur while the Fe₂O₃ is only reduced to Fe₃O₄. Separation of Fe and Mn can be realized by leaching the roasted products in a dilute sulphuric acid solution. Leaching efficiencies of 95.6% for Mn and only 14.5% for Fe are achieved (Zhang et al., 2013; You et al., 2017). In the MnO₂-Fe₂O₃-SO₂-O₂ system (Bolsaitis and Nagata, 1980; Ye et al., 2013; You et al., 2015), MnO₂ is readily transformed to MnSO₄. However, the sulfation of ferric oxide only occurs in the presence of both SO₂ and O₂. Controlling the MnO₂-Fe₂O₃-SO₂-O₂ system in the thermodynamic stability region for MnSO₄ and Fe₂O₃ can realize the selective sulfation of MnO₂. Afterwards, water leaching can realize the separation of Mn and Fe. Experimental results for sulfation roasting and water leaching reveal that 90.6% of manganese and only 3.5% of iron are extracted when the sulfation roasting test is conducted at 500 °C for 60 min with SO₂ partial pressure (SO₂/(SO₂ + N₂)) of 0.5–1.0 vol% (You et al., 2015). The detailed sulfation reduction and leaching conditions of the Fe-Mn ores are listed in Table 6.

A demonstration plant for the sulfation roasting with sulfur followed by acid leaching of Fe-Mn ores was set up successfully in Jinxu Metallurgy and Chemistry Company, Hunan province, China. Fig. 11

Table 6
Sulfation roasting-leaching results of different Fe-Mn ores.

Main minerals	Chemical compositions (wt.%)			Reduction conditions			Leaching conditions			Leaching efficiency (wt.)		Ref.
	Fe	Mn	SiO ₂	T (°C)	Time (min)	Reductant				Fe	Mn	
Manganese dioxide, Hematite, Quartz	5.68	21.43	36.36	680	70	160% sulfuric acid, 16% biogas residual	Water	1.0 mol/L H ₂ SO ₄ , 25 °C, S/L = 5:1, 5 min	~100 ^a	~100		Cai et al. (2012)
Lithiophorite, hematite, quartz	10.69	21.10	31.62	550	10	Sulfur (S/Mn mole ratio 0.5:1)			14.50	95.60		Zhang et al. (2013)
Cryptomelane, goethite, hematite, quartz	14.30	30.35	18.63	500	60	1 vol.%SO ₂ + 99 vol.%N ₂	Water, 50 °C, S/L = 5:1, 15 min		3.8	90.60		You et al. (2015)
Pyrolusite, todorokite, hematite, goethite, quartz	28.70	21.97	–	500	60	1 vol.%SO ₂ + 99 vol.%N ₂	Water, 50 °C, S/L = 5:1, 15 min		3.2	85.2		You et al. (2015)
Manjirolite, goethite, quartz	20.95	19.44	–	500	60	1 vol.%SO ₂ + 99 vol.%N ₂	Water, 50 °C, S/L = 5:1, 15 min		2.8	91.2		You et al. (2015)
Manjirolite, goethite, quartz	14.30	30.35	18.63	400–600	30	Sulfur(S/Mn mole ratio 0.6:1)	1.0 mol/L H ₂ SO ₄ , 25 °C, S/L = 5:1, 10 min		< 10.0	> 95.0		You et al. (2017)
Pyrolusite, Hematite, Quartz	8.20	40.10	31.62	850–950	~30	Sulfur (sulfur/ore mass ratio of 1:3)	1.0 mol/L H ₂ SO ₄ , 25 °C, S/L = 5:1, 5–10 min		< 10.0	96–98.2		Jinxu Company, 2012 ^b

^a The Fe leaching efficiency is not considered by Cai et al. (2012), however, the Fe leaching efficiency is estimated to be 100% since the ore is roasted with concentrated H₂SO₄ at 680 °C. All the iron-bearing minerals are dissolved in molten H₂SO₄.

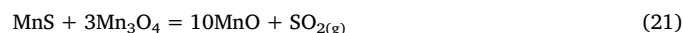
^b Industrial test is conducted in a fluidized bed furnace in 2012 at Jinxu Metallurgy and Chemistry Company, Hunan province, China.

shows the flow sheet for industrial sulfation roasting followed by acid leaching of the Fe-Mn ores, producing MnSO₄ and concentrated H₂SO₄. Without building new production line, this process combines the traditional process for the production of concentrated H₂SO₄ and the ore leaching process. During the development of this new process, a lot of creative technologies and apparatuses were invented (Jiang et al., 2010, 2012a, 2012b; Liu and Liu, 2010; Liu, 2011, 2012a, 2012b, 2012c, 2012d; Zhang et al., 2012b).

The main roasting equipment for the sulfation roasting is the fluidized bed furnace. The sulfur and Fe-Mn ores mixtures with mass ratio of 1:3 (sulfur/ore) are charged into the furnace and then roasted at 850–900 °C. During the roasting process, the sulfur is combusted to generate heat energy and gaseous SO₂. Simultaneously, the MnO₂ is reduced to MnO or MnSO₄ by the gaseous sulfur and SO₂, while the Fe₂O₃ is only reduced to Fe₃O₄. The upward flowing gas containing 10–13 vol% SO₂ is handled in turn by gas purification, contact oxidation in a chamber with catalysts, and absorption by 98% H₂SO₄ to produce concentrated H₂SO₄. On the other hand, the roasted Fe-Mn ores are discharged from the fluidized bed furnace via a sealed cooling and discharging system to the acid leaching tank. MnO and MnSO₄ dissolve into the dilute H₂SO₄ solution while little iron oxides particles dissolve in the acid solution, and most of the iron oxides are discharged as the Fe-rich residual for the subsequent recovery of iron. During the industrial tests, the reduction degree of MnO₂ to MnO and MnSO₄ can reach 96–98.2%. Simultaneously, qualified concentrated H₂SO₄ from the SO₂ flue gas and electrolytic metal manganese (EMM) from the as-prepared MnSO₄ solution are produced.

An electrolytic manganese factory with a capacity of 10, 000 tons EMM products per year was built in Jinxu Metallurgy and Chemistry Company, Hunan, China. As estimated, with the construction, implementation and refinement of the production line, more than half of the Mn-bearing slag and waste water discharge were reduced (Jinxu Company, 2012).

Sulfation roasting with sulphur or SO₂ displays better selective reduction of MnO₂ and Fe₂O₃. As MnS forms in the MnO₂-Fe₂O₃-S system, poisonous H₂S gas is inevitably generated during the following leaching process according to Reaction (20), resulting in serious environmental pollution. However, when Mn₃O₄ and MnS are coexisting in the aqueous solution, the formation of H₂S can be prevented according to Reaction (21). Sulfation roasting with SO₂ is found as a promising technique for the separation of Mn and Fe from the Fe-Mn ores. The main product is MnSO₄. It's notable that MnS is not generated during the roasting process and thus the environmental problem of H₂S generation is avoided.



3.6. Others

The combined processes of chemical and multiple physical beneficiation methods have also been applied for the separation of Mn and Fe (Dai and Zhao, 2008; Zhan et al., 2013). Dai and Zhao 2008 firstly investigated the magnetic reduction roasting and magnetic separation of Fe-Mn ores from Yunnan, China. The Fe grade is increased from 27.82% for raw Fe-Mn ores to 55.50% in the magnetic concentrate with a recovery of 65.81%. After that, reverse flotation of the non-magnetic products was conducted to further enrich the Mn grade. The Mn grade was enhanced from 18.80% in the non-magnetic products to 34.55% in the manganese concentrates with a recovery of 78.47% after reverse flotation. Zhan et al. (2013) also reported a combined technology including washing, magnetic reduction roasting followed by low intensity magnetic separation, and then high intensity magnetic separation and gravity separation were applied to further upgrade the Mn and Fe grades of the intermediate beneficiation products.

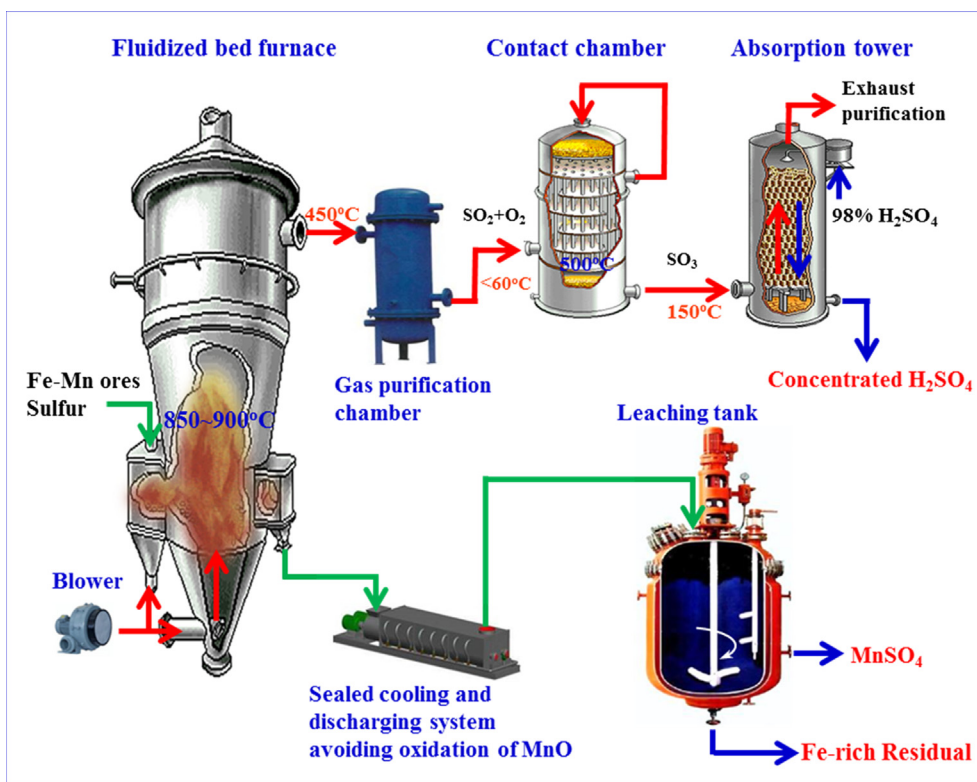


Fig. 11. Flow sheet of industrial sulfation roasting with sulfur followed by leaching of Fe-Mn ores for producing MnSO₄ and concentrated H₂SO₄ (Jiang et al., 2010; Liu and Liu, 2010; Liu, 2011, 2012a, 2012b, 2012c, 2012d).

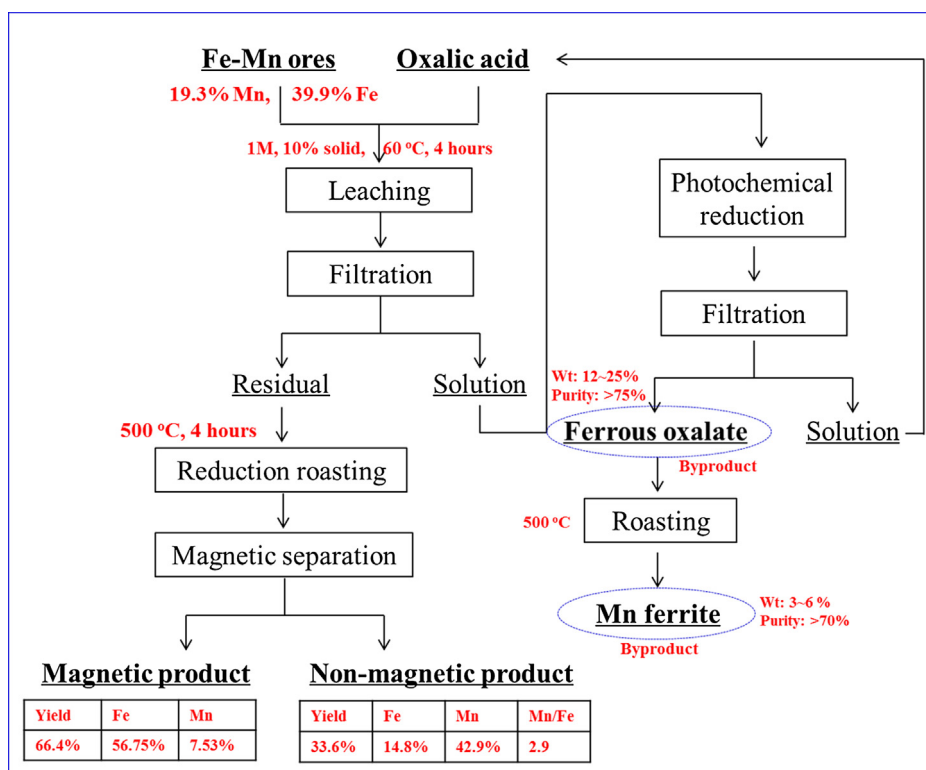


Fig. 12. A physicochemical beneficiation process for Fe-Mn ores (Singh and Biswas, 2017).

Fig. 12 displays a new and cost effective physicochemical beneficiation process for the Fe-Mn ores to recover the valuable iron-bearing by-products (iron oxalate or manganese ferrite) and manganese-rich concentrate (Singh and Biswas, 2017). It's found that 1 mol/L oxalic

acid can leach five times more Fe than Mn from Fe-Mn ores when the ores are leached at 60 °C for 2 h. Photoreduction of the leached solution with sunlight provides ferrous oxalate dihydrate and the dihydrate can be further converted into manganese ferrite by roasting at 500 °C for

4 h. On the other hand, undissolved manganese and iron in the leaching residues can be further recovered by the roasting and magnetic separation method. This proposed process can recover 0.12–0.25 tons of iron oxalate and 0.33–0.50 tons of manganese concentrate (Mn: 42–55% and Mn/Fe: 2.91–4.5) by processing 1 ton of Fe-Mn ores (Mn: 19.3–31.3% and Mn/Fe: 0.48–1.16).

Based on our previous investigations on the utilization of Fe-Mn ores and behaviors of Fe_2O_3 - MnO_2 system roasted under different atmospheres, we found that manganese ferrites were readily formed and they have strong magnetism (Zhang et al., 2016b; Liu et al., 2017c, 2017d, 2018b). In light of the discovery, a novel calcified roasting followed by magnetic separation process was proposed by the authors' group (Zhang et al., 2017b) to deal with Fe-Mn ores in which manganese ferrites were easily formed from manganese and iron oxides under a controlled atmosphere. Then, high-purity manganese ferrites are recovered by magnetic separation, and the products can be used as a precursor for the preparation of soft magnetic materials because they have been tested to have strong saturation magnetization and low magnetic coercive force (Liu et al., 2017c, 2017d). The gangue minerals in the Fe-Mn ores are removed effectively from the magnetic manganese ferrites by multistage magnetic separations. Meanwhile, the recovery ratios of Fe and Mn can reach 80–90%. This process is especially suitable for handling the Fe-Mn ores with a high proportion of Fe-Mn composite oxides (as shown in Fig. 2). The process can provide an alternative and effective method to synchronously recover Fe and Mn elements from the Fe-Mn ores, and can produce high added value final magnetic products (manganese ferrites). We are currently doing further research on the large-scale utilization of the Fe-Mn ores by this new process.

4. Conclusions

90–95% of mined manganese is utilized in the Iron & Steel industry to produce Mn alloys. In 2016, the global manganese mine production was 16 Mts and the global output of Mn alloys totaled 17.2 Mts. Chinese manganese mine production only accounted for 16.9–18.8% of the global mine production while Chinese manganese smelters produced 55% of the global Mn alloy total in 2016. Hence, it's imperative to exploit and utilize the abundant Fe-Mn ore resources available in China to ensure the sustainable development of manganese industry.

Numerous investigations have been conducted on the extraction and separation of Mn and Fe from Fe-Mn ores. Physical separation including washing, gravity, magnetic and flotation, direct reduction leaching, Mn rich slag smelting, carbothermic reduction roasting followed by magnetic separation or acid leaching, sulfation roasting followed by acid leaching or water leaching, as well as various combinations of the aforesaid processes have been previously applied to deal with Fe-Mn ores. The key point for the extraction and separation of Mn and Fe is to enlarge the magnetic, solubility and fusibility differences among Mn, Fe and gangue constituents.

Physical separation techniques have a relatively inferior separation effect on Mn and Fe due to the similar physical properties of the manganese and iron minerals as well as the complex mineralogy of many Fe-Mn ores. Direct reduction leaching of Fe-Mn ores with oxalic acid and methanol presents better selective leaching of manganese and iron. However, it's uneconomical owing to the high price of organic reductants and the high leaching temperature. Even though the Mn-rich slag smelting method is characterized as having a large production and high efficiency, it has defects due to its energy-intensive consumption and heavy pollution. Carbothermic reduction roasting followed by separation process shows a superior separation effect for Mn and Fe from the Fe-Mn ores. In the direct reduction roasting process, alkaline additives can facilitate the growth of metallic iron and destroy the structure of difficult to reduce silicates to therefore improve the reduction of iron and manganese oxides. Sulfation roasting with SO_2 or sulfur followed by leaching is a prospective process for the selective

extraction of Mn and Fe from Fe-Mn ores because the combination reactions of MnO_2 and S (or SO_2) forming MnO (or MnSO_4) are more thermodynamically favorable. Combined processes of chemical and multiple physical beneficiation methods are recommended to separate Mn and Fe from Fe-Mn ores to achieve high-quality Fe-rich products and Mn-rich products.

A calcified roasting followed by magnetic separation process is proposed to synchronously recover manganese and iron from Fe-Mn ores. Precursor magnetic materials, high-quality manganese ferrites with strong saturation magnetization and low magnetic coercive force, can be produced by this process. A large-scale test is being conducted to further identify the overall effectiveness.

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